

Lifecycle Wage Dynamics in the United States: How the Gender Gap and College Premium Evolve with Age

ECO 340 Data Project

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Data: IPUMS CPS, 2012 March ASEC Supplement

1. Introduction

A foundational insight of labour economics is that wages are not static but evolve systematically over a worker's career. The classical human capital framework, developed by Mincer (1974) and Becker (1964), predicts that wages rise with age as workers accumulate on-the-job experience, and that more-educated workers earn persistently higher wages because education raises productivity. Yet these aggregate predictions mask substantial heterogeneity: the pace at which wages grow over the lifecycle differs by gender and by education level, and the gaps between groups are themselves not fixed but widen, narrow, or shift in shape across the career.

This paper uses the 2012 March Annual Social and Economic (ASEC) Supplement of the Current Population Survey (CPS) to investigate how the gender wage gap and the college premium vary with age. The central research questions are: (1) Do men and women experience different rates of wage growth over the lifecycle, and does this differ by education level? (2) Does the gender wage gap widen or narrow with age, and is this pattern uniform across education groups? (3) Does the return to a college degree grow over the working life, and does this lifecycle profile differ by gender?

These questions bear directly on the economics of earnings inequality. A gender gap that widens with age is consistent with theories emphasising career interruptions due to childbearing, occupational segregation, or discriminatory promotion practices that compound over time (Goldin, 2014; Bertrand, Goldin, and Katz, 2010). A college premium that grows with age is consistent with signalling models (Spence, 1973) in which the credential permits access to jobs with steeper wage-tenure profiles, and with human capital models predicting higher returns to on-the-job learning for college graduates.

Using a baseline Mincer regression with age, education, gender, and state fixed effects, and an expanded specification with age-education, age-gender, and gender-education interaction terms, we document three main findings. First, men enjoy a substantially larger seniority premium than women at every education level. Second, the gender wage gap is largest in mid-career, peaking near age 42 for college-educated workers. Third, the college premium grows significantly over the lifecycle for both sexes, and women enjoy a modestly larger college premium than men at every age despite facing a larger aggregate gender gap. A robustness check replacing annual wage with hourly wage reveals that roughly one-third of the annual gender gap reflects differences in hours worked rather than in wage rates per se.

2. Data

2.1 Source and Variables

The data come from the 2012 March ASEC Supplement of the Current Population Survey (CPS), accessed via the IPUMS platform at the University of Minnesota (Flood et al., 2023). The CPS is a monthly household survey sponsored by the U.S. Census Bureau and the Bureau of Labor Statistics; the March ASEC wave contains detailed annual income, hours, and demographic information for the prior calendar year (2011). We use the following variables: age, sex, educational attainment, state of residence (STATEFIP), annual wage and salary income (INCWAGE), usual weekly hours worked (UHRSWORKLY), and weeks worked in the prior year (WKSWORK1).

2.2 Sample Selection and Cleaning

We apply a sequence of sample selection criteria. First, we restrict to workers aged 25 to 60, excluding students and early retirees. Second, we drop observations with missing values on any key variable. Third, we drop individuals with INCWAGE below \$2,200, roughly one month of

full-time work at the 2011 federal minimum wage. Fourth, we drop individuals whose implied hourly wage exceeds \$450 while annual hours are below 100, as these likely reflect data entry errors. Finally, to reduce the influence of extreme outliers, we trim the top and bottom 1% of INCWAGE. The resulting sample contains 69,939 individuals.

Table 1: Summary Statistics for Raw, Cleaned, and Trimmed Samples

Sample	N	Mean Age	SD Age	Mean Wage	SD Wage	Mean Hrs/Wk	Skew	Kurt	IQR Wage
Raw	201,398	35.36	22.27	\$23.2M	42.2M	63.1	1.26	2.61	\$71
Cleaned	71,298	42.01	9.87	\$50,439	60,318	89.5	8.35	116	\$38,000
Trimmed 1%	69,939	42.00	9.87	\$46,910	35,993	89.4	3.11	8.00	\$38,000

Note: INCWAGE statistics in U.S. dollars. The dramatic reduction in mean INCWAGE from raw to cleaned reflects removal of topcoded and erroneous values. The further reduction in standard deviation and kurtosis from cleaned to trimmed reflects removal of top- and bottom-1% earners.

2.3 Education Classification

Educational attainment in the CPS is recorded as a detailed categorical variable with 17 codes. For the expanded interaction model, we collapse these into five groups: less than high school diploma (reference category), high school diploma (EDUC = 73), some college or associate degree (EDUC in {81, 91, 92, 111}), bachelor's degree (EDUC = 123), and graduate or professional degree (EDUC in {124, 125}). This grouping preserves the major steps in the U.S. educational ladder while maintaining adequate cell sizes for interaction estimation.

3. Empirical Strategy

3.1 Baseline Mincer Regression

Our first model is a cross-sectional Mincer regression of log annual wages on a full set of age dummies, education dummies, a female indicator, and state fixed effects. Age 25 is the reference

category for age dummies, less-than-high-school is the reference for education, and male is the reference for gender. The coefficient on Female thus measures the average log wage gap between women and men conditional on age, education, and state. This specification imposes the restriction that the gender gap and education returns are constant across the age range, which we relax in the second model.

3.2 Expanded Mincer with Interaction Terms

Our second model relaxes additivity by including pairwise interaction terms between age, education, and gender. This specification allows the gender gap to vary with both age and education, and the seniority premium (wage growth between ages 30 and 55) to differ by education group and by sex. We use predictions from this richer model to trace the lifecycle paths of the gender gap and the college premium, constructing hypothetical individuals with specified age-sex-education combinations at the baseline (modal) state.

3.3 Robustness Check: Hourly Wages

Annual wage income conflates the hourly wage rate with hours of work. Since women work on average fewer annual hours than men, partly due to part-time arrangements and career interruptions, the annual gender gap mechanically overstates the gap in hourly compensation. As a robustness check, we re-estimate the baseline Mincer specification replacing log annual wage with log hourly wage, constructed as INCWAGE divided by the product of usual weekly hours and weeks worked. Comparing the two sets of results allows us to decompose the annual gender gap into an hours component and a rate component.

4. Results

4.1 Descriptive Patterns

Figure 1 shows mean log annual wage by age for men and women separately, broken down by educational attainment. Several patterns are immediately apparent. Wages rise with age for all groups up to approximately age 50-55, consistent with the human capital model. The wage hierarchy by education is steep and consistent: graduate degree holders earn substantially more than college graduates, who in turn earn more than those with only some college or a high school diploma. The slopes of the age-wage profiles differ across education groups, with college and graduate-educated workers showing steeper profiles. Finally, women's wage profiles lie consistently below men's at every age and education level, but the gap varies in size across the lifecycle.

Figure 1: Age-Wage Profiles by Education and Gender
(CPS 2012)

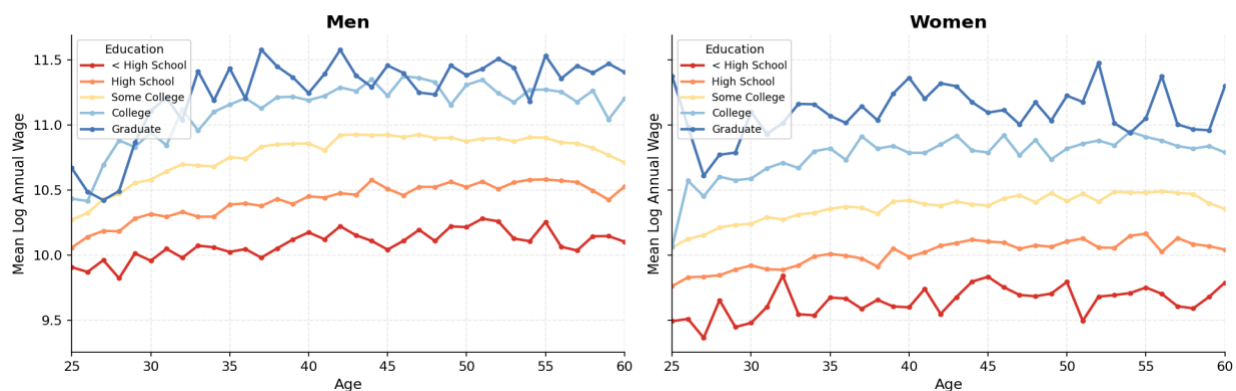


Figure 1: Age-Wage Profiles by Education and Gender (CPS 2012)

Figure 4 shows the full distribution of log wages by education level using violin plots. Higher education is associated not only with higher mean wages but also with lower variance and greater symmetry, consistent with the view that education provides a wage floor by opening access to professional and managerial occupations. The distribution for the less-than-high-school group is notably left-skewed, reflecting concentration at very low wage levels.

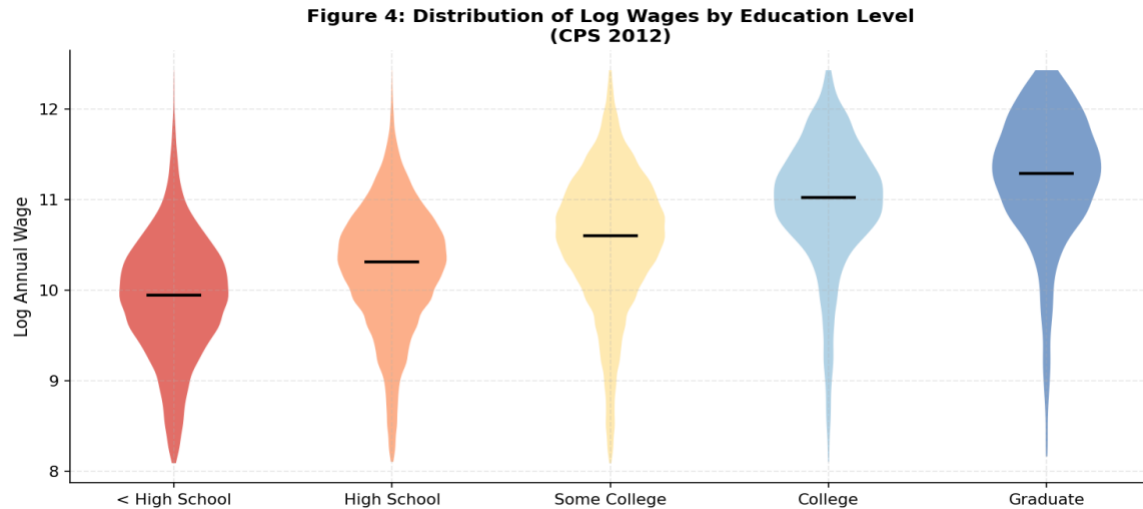


Figure 4: Distribution of Log Wages by Education Level (CPS 2012)

4.2 Baseline Mincer Regression

Table 2 reports selected coefficients from the baseline Mincer regression. All age, gender, and higher-education coefficients are statistically significant at the 0.1% level.

Table 2: Baseline Mincer Regression Results (Selected Coefficients)

Variable	Estimate	Std. Error	t-stat	p-value
Age 30 (vs. 25)	0.2272	0.0223	10.21	<0.001
Age 35	0.3753	0.0222	16.93	<0.001
Age 40	0.4221	0.0220	19.22	<0.001
Age 45	0.4561	0.0220	20.74	<0.001
Age 50	0.4755	0.0217	21.92	<0.001
Age 55	0.5210	0.0227	22.99	<0.001
High School (vs. <HS)	0.09 to 0.46	varies	—	varies
College degree (EDUC=123)	1.1934	0.0603	19.78	<0.001
Graduate degree	1.37 to 1.45	varies	—	<0.001
Female (vs. Male)	-0.4059	0.0051	-79.87	<0.001
R-squared	0.2632			
N	69,939			

Note: Dependent variable is log annual wage income. Reference categories: age 25, less-than-high-school education, male. State dummies included but not reported. High School and Graduate rows show ranges across component dummies. Standard errors are heteroskedasticity-robust.

The age dummies display a classic concave wage-experience profile. The wage gain between age 25 and age 30 is 22.7 log points (approximately 25.5%), rising to 52.1 log points at age 55. The seniority gap between age 59 and age 25, computed from the full regression, is approximately 40.5 log points, implying wages roughly double over this age range holding all else equal.

The gender coefficient of -0.406 implies women earn approximately 33% less than men in annual wages, conditional on age, education, and state. This large and highly significant gap is consistent with the well-documented U.S. gender wage gap literature. The college premium relative to less-than-high-school is 1.193 log points, while the graduate degree premium reaches 1.37-1.45 log points. Geographic variation is substantial: the District of Columbia commands a 15.9 log point premium over the baseline state, while Montana sits 14.9 log points below, likely reflecting differences in cost of living, industrial composition, and local labour market conditions.

4.3 Seniority Gaps by Education and Gender

Table 3 reports predicted log wage gains between age 30 and age 55 from the expanded interaction model, by education and sex. This seniority gap measures the steepness of the lifecycle wage profile.

Table 3: Seniority Gap (Predicted Log Wage at Age 55 minus Age 30) by Education and Gender

Education	Men	Women	Difference (Women minus Men)
< High School	0.305	0.238	-0.067
High School	0.288	0.221	-0.067
Some College	0.317	0.250	-0.067

College	0.382	0.314	-0.068
Graduate	0.255	0.188	-0.067

Two striking patterns emerge. First, at every education level men experience a larger seniority premium than women by approximately 6.7 log points, a difference that is almost exactly constant across education groups. This implies that men's wages grow faster over the career, widening the raw gender gap as workers age. Second, the seniority gap is highest for college-educated workers (0.382 for men, 0.314 for women) and lowest for graduate degree holders (0.255 for men, 0.188 for women). This non-monotone pattern is economically plausible: graduate degree holders start at higher base wages after longer schooling, leaving less room for further growth, while college graduates may benefit more steeply from employer learning and internal promotion opportunities. Figure 6 visualises the predicted profiles for all five education groups by gender.

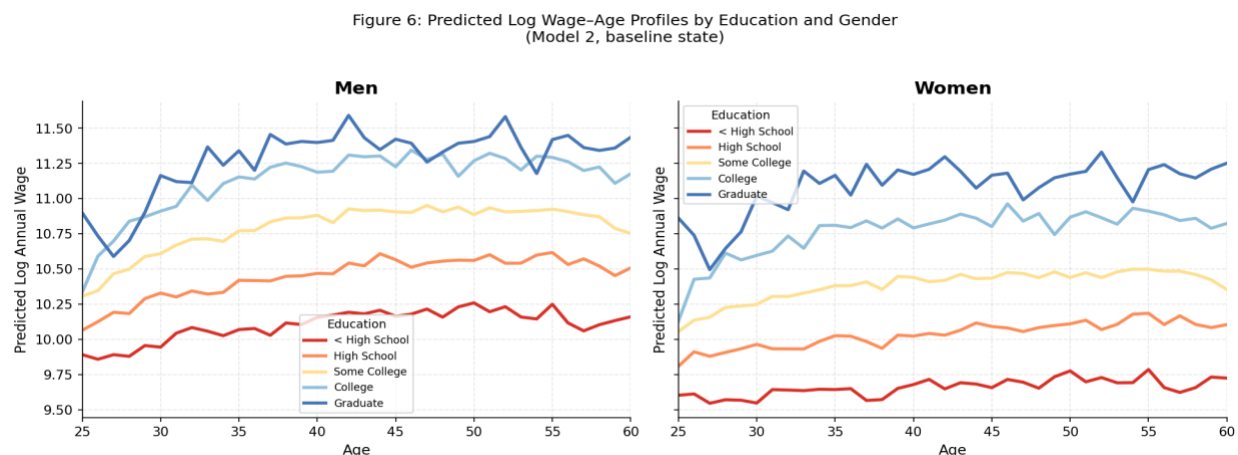


Figure 6: Predicted Log Wage-Age Profiles by Education and Gender (Expanded Model)

4.4 The Gender Wage Gap over the Lifecycle

Figure 2 shows the raw gender log wage gap (women minus men) by age, separately for each education group. The gap is universally negative, but its magnitude varies substantially with age. For most education groups the gap is smallest in the mid-to-late 20s, widens through the 30s and early 40s, and then partially narrows again in the 50s. This inverted-U pattern in absolute value is

consistent with explanations emphasising childbearing and family formation in the 30s as key drivers of wage divergence between men and women.

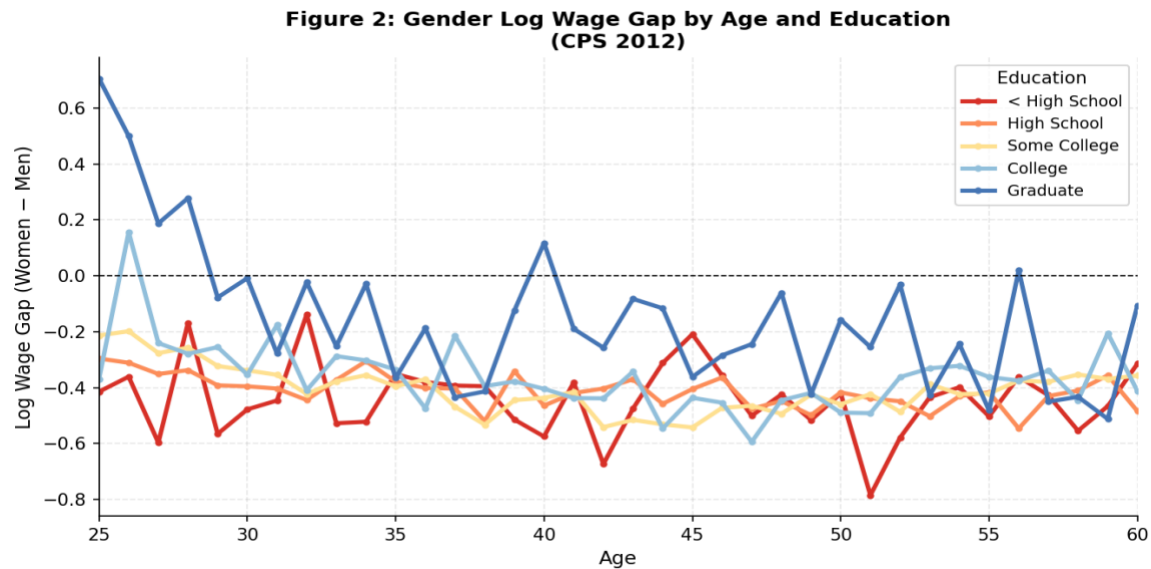


Figure 2: Gender Log Wage Gap by Age and Education Level (CPS 2012)

Table 4 quantifies this pattern for college-educated workers from the interaction model. At age 28 the gender gap is -0.226 log points (approximately 20% lower wages for women). By age 42 the gap has more than doubled to -0.462 log points (approximately 37% lower). By age 55, the gap partially reverses to -0.382 log points. The peak gap in mid-career is consistent with Bertrand et al. (2010) and Goldin (2014), who document that the gender earnings gap among highly educated professionals widens sharply in the 30s and early 40s, driven in large part by the penalty associated with reduced temporal flexibility and career interruptions.

Table 4: Predicted Gender Wage Gap Among College-Educated Workers at Selected Ages

Age	Gender Gap (Women minus Men, log wage)	Implied % Difference
28	-0.226	-20.2%
42	-0.462	-37.0%
55	-0.382	-31.8%

Note: Gap is predicted log wage for women minus predicted log wage for men, both with college education at the baseline state. Implied percentage differences computed as $\exp(\text{gap})$ minus 1.

4.5 The College Premium over the Lifecycle

Figure 3 plots the raw college premium (mean log wage for college graduates minus mean log wage for high school graduates) by age, separately for men and women. The premium rises steeply over the lifecycle, from roughly 0.4-0.5 log points in the late 20s to approximately 0.8-1.0 log points by age 55. This growing lifecycle premium is consistent with signalling models (Spence, 1973) in which a college degree unlocks access to occupations with steeper wage-tenure profiles, and with human capital models in which college-educated workers have higher returns to on-the-job learning.

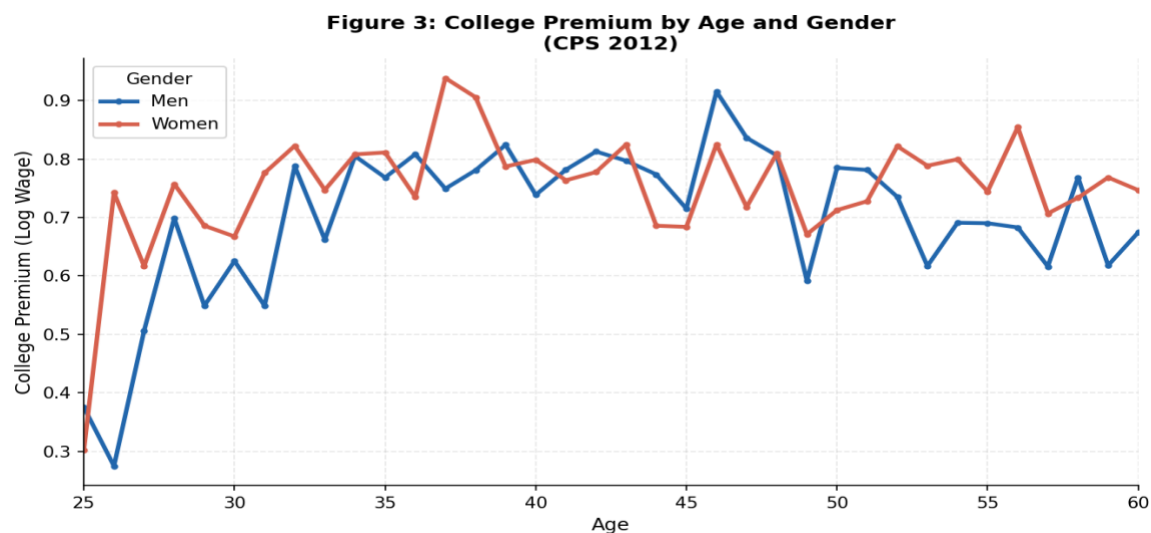


Figure 3: College Premium by Age and Gender (Raw Means, CPS 2012)

Table 5 quantifies the interaction model predictions. At age 26 the college premium is 0.465 log points for men (approximately 59% higher wages) and 0.515 log points for women (approximately 67%). By age 48 these premiums grow to 0.760 and 0.810 respectively. Women enjoy a consistently higher college premium than men at both ages by approximately 5 log points. This pattern is consistent with the equalising role of education in the literature: while education does

not eliminate the gender gap, it narrows it relative to what women would earn without the degree, and this gap-narrowing effect grows over the career.

Table 5: Predicted College Premium (College vs. High School) by Gender at Selected Ages

Age	Men (log)	Women (log)	Men (approx %)	Women (approx %)
26	0.465	0.515	59.2%	67.4%
48	0.760	0.810	113.8%	124.8%

Note: Premium is predicted log wage for a college graduate minus predicted log wage for a high school graduate at the baseline state. Implied percentage premiums approximated as $\exp(\log \text{premium})$ minus 1.

4.6 Robustness: Hourly vs. Annual Wages

A natural concern with results based on annual wage income is that women work fewer hours on average, so the annual gender gap partly reflects an hours gap rather than a wage rate gap. Table 6 compares the gender gap from the baseline Mincer model using annual wages with the analogous estimate using log hourly wages.

Table 6: Robustness Check: Gender Wage Gap Using Annual vs. Hourly Wages

Wage Measure	Gender Gap (log)	Implied % Gap	R-sq	N
Annual wage income	-0.406	-33.4%	0.263	69,939
Hourly wage rate	-0.249	-22.0%	0.232	69,939

The hourly gender gap of -0.249 is substantially smaller in absolute value than the annual gap of -0.406. Using the approximation $\exp(\text{gap})$ minus 1, the annual gap implies women earn approximately 33% less per year, while the hourly gap implies approximately 22% less per hour. The difference of roughly 11 percentage points represents the portion of the annual gap attributable to women working fewer annual hours. Figure 5 visualises the two series across the age distribution. Both show a similar mid-career widening pattern, confirming that the lifecycle pattern

we document is not purely an artefact of changing hours patterns with age but reflects genuine differences in wage rates.

Figure 5: Robustness Check – Annual vs. Hourly Wage Gender Gap (CPS 2012)

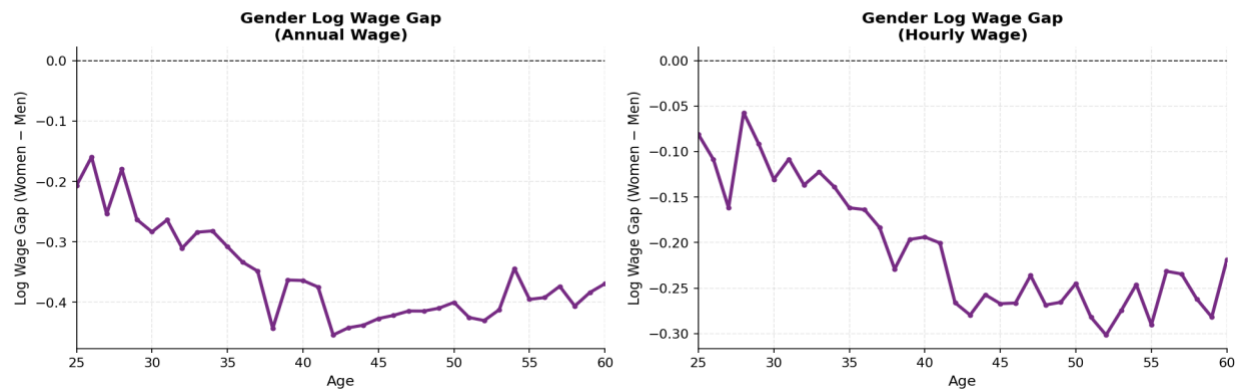


Figure 5: Robustness Check: Annual vs. Hourly Gender Wage Gap by Age (CPS 2012)

5. Discussion and Limitations

5.1 Interpretation of Findings

Our three main findings can be interpreted through complementary theoretical lenses. The larger male seniority premium is consistent with statistical discrimination models (Phelps, 1972; Arrow, 1973), in which employers who anticipate higher female quit rates due to family formation invest less in firm-specific training for women, generating slower wage growth. It is also consistent with household specialisation models (Becker, 1985), in which married women reduce labour supply intensity in response to childbearing, slowing on-the-job human capital accumulation relative to men.

The peak gender gap in mid-career aligns closely with Goldin's (2014) insight that in high-skill occupations such as law, finance, and consulting, compensation is highly convex in hours worked. Because women in their 30s and early 40s are more likely to reduce hours or take career interruptions for family reasons, they face disproportionate wage penalties that widen the gap. The partial narrowing of the gap after age 50 may reflect cohort effects, or selection effects whereby

women with lower wages are more likely to exit the workforce, raising the observed mean among those who remain.

The growing college premium and its larger size for women are consistent with sheepskin effects (Jaeger and Page, 1996) and the signalling model: the value of the credential is realised progressively through career advancement and employer learning. Women's larger premium may reflect the fact that, among less-educated workers, women face especially severe wage penalties due to occupational segregation into low-wage service jobs, so a college degree provides a larger escape premium for women than for men.

5.2 Limitations

This analysis has several important limitations. First and most fundamentally, a cross-sectional analysis cannot distinguish age effects from cohort effects. Workers aged 55 in 2012 entered the labour market around 1982 under very different economic and social conditions than those aged 25 in 2012. The lifecycle profiles we observe may partly reflect generational differences rather than pure wage growth for any given cohort. Panel data such as the NLSY or PSID would be required to cleanly identify age effects.

Second, OLS estimation of the Mincer equation is subject to well-known biases. Ability bias arises because more-educated workers may be intrinsically more productive, so education coefficients overstate the causal return to schooling. The 2012 CPS does not contain direct ability measures such as ASVAB scores, so we cannot address this bias. Additionally, measurement error in reported hours worked may attenuate the hourly wage estimates toward zero, partly explaining the smaller hourly gender gap relative to the annual gap.

Third, our sample selection criteria may introduce selection bias. Workers with very low earnings and those who have exited the workforce entirely are excluded, and the propensity to be excluded is correlated with gender and education. This is particularly important for interpreting the gender gap results, since women have lower labour force participation rates and are more likely to report zero earnings in any given year.

Fourth, the CPS does not contain information on occupation, industry, or firm characteristics. A large body of research (Blau and Kahn, 2017) demonstrates that occupational and industry segregation account for a substantial share of the gender wage gap. Our estimates therefore capture a mix of within-occupation and between-occupation effects that we cannot disentangle with available data.

5.3 Future Work

Several extensions would be valuable. Using panel data such as the NLSY or PSID would allow us to follow actual workers over time, distinguishing age from cohort effects and enabling individual fixed effects to control for unobserved ability. Combining the CPS with occupational data from O*Net would allow direct tests of the Goldin (2014) hypothesis about hours flexibility by controlling for the degree of temporal flexibility required by different occupations. Extending the analysis to multiple CPS years would allow examination of whether the lifecycle patterns we document have changed over time in response to rising women's labour force participation and the gradual closing of the gender gap. Finally, a regression discontinuity design exploiting parental leave or childcare benefit age thresholds could more cleanly identify the causal impact of family formation on the lifecycle gender gap.

6. Conclusion

Using the 2012 March CPS and a Mincer wage framework with interaction terms, this paper documents three key findings about how the gender wage gap and the college premium vary over the working lifecycle. Men's wages grow faster with age than women's at every education level, widening the gender gap as workers age. The gender gap among college-educated workers peaks in mid-career at roughly 37% at age 42, consistent with the literature emphasising the role of family formation and hours constraints. The college premium grows substantially over the lifecycle and is modestly larger for women than for men, suggesting that education partially offsets but does not eliminate the gender wage penalty. A robustness check using hourly rather than annual wages confirms that roughly one-third of the annual gender gap reflects differences in hours worked, while two-thirds reflects differences in wage rates.

These findings add to the literature on lifecycle earnings dynamics and earnings inequality, and raise important policy questions about the mechanisms that sustain the gender wage gap even among highly educated workers. While the cross-sectional design limits causal inference, the patterns are consistent with and provide CPS-based evidence supporting the insights of Goldin (2014) and Bertrand et al. (2010) about the importance of temporal flexibility in high-skill occupations for explaining persistent gender wage inequality.

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